

# Differential Equations

An MIT-Style Course with Derivations, Proofs, Geometry, and  
Applications

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From first-order ODEs to systems, Laplace transforms, numerical methods, stability, Fourier  
modes, and PDEs

**How to use these notes.** Read each chapter in order. Do not memorize formulas before understanding the mechanism that creates them. The proofs are deliberately chosen to explain *why* the standard methods work.

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## Course roadmap

Differential equations are the mathematics of **change**. A differential equation does not usually tell you the value of a quantity directly; it tells you how the quantity changes, and your task is to reconstruct the whole function from that local rule.

### Core idea

The recurring pattern is

$$\text{local law of change} \implies \text{global motion or shape.}$$

Examples include Newton's law, population growth, electric circuits, heat flow, waves, feedback systems, and fluid models.

The course is organized into twelve chapters:

1. Language, geometry, and initial-value problems.
2. Separable and autonomous equations.
3. First-order linear, exact, and Bernoulli equations.
4. Existence, uniqueness, Picard iteration, and Gronwall's inequality.
5. Second-order linear equations and characteristic roots.
6. Oscillations, damping, forcing, and resonance.
7. Linear operators, Wronskians, and variation of parameters.
8. Systems, eigenvalues, matrix exponentials, and phase portraits.
9. Nonlinear systems, linearization, and Lyapunov stability.
10. Laplace transforms, step inputs, impulses, and convolution.
11. Power series and numerical methods.
12. Boundary-value problems, Fourier series, and a bridge to PDEs.

### Minimal prerequisites

You should be comfortable with derivatives, the chain rule, basic integration, Taylor expansion, complex numbers, and elementary matrices/eigenvalues. When a result depends on a deeper theorem, the relevant idea is stated explicitly.

# 1 What a differential equation really says

## 1.1 From values to laws of change

Suppose  $y(t)$  measures a quantity evolving with time. Its derivative

$$y'(t) = \frac{dy}{dt}$$

measures its instantaneous rate of change. A differential equation relates  $y$  to one or more derivatives.

The simplest important model is

$$y' = ky.$$

This means the rate of change is proportional to the current amount. Because

$$\frac{d}{dt}e^{kt} = ke^{kt},$$

the solutions are

$$y(t) = Ce^{kt}.$$

An initial condition such as  $y(0) = 5$  gives  $C = 5$  and therefore  $y(t) = 5e^{kt}$ .

## 1.2 Order, linearity, and initial data

The **order** is the highest derivative appearing. For example,

$$y'' + 3y' + 2y = \sin t$$

is second order.

A differential equation is **linear** when the unknown function and its derivatives occur linearly. Thus

$$y'' + ty' + \sin(t)y = e^t$$

is linear, while

$$yy'' + (y')^2 = 0$$

and

$$y' + y^2 = 0$$

are nonlinear.

A first-order equation normally needs one initial condition,

$$y' = f(t, y), \quad y(t_0) = y_0,$$

while a second-order equation normally needs two,

$$y'' = f(t, y, y'), \quad y(t_0) = y_0, \quad y'(t_0) = v_0.$$

This mirrors mechanics: acceleration alone does not determine motion unless initial position and velocity are known.

## 1.3 Direction fields: the geometry before the algebra

For

$$y' = f(x, y),$$

each point  $(x, y)$  is assigned a slope  $f(x, y)$ . A solution curve is one whose tangent slope matches that field everywhere.

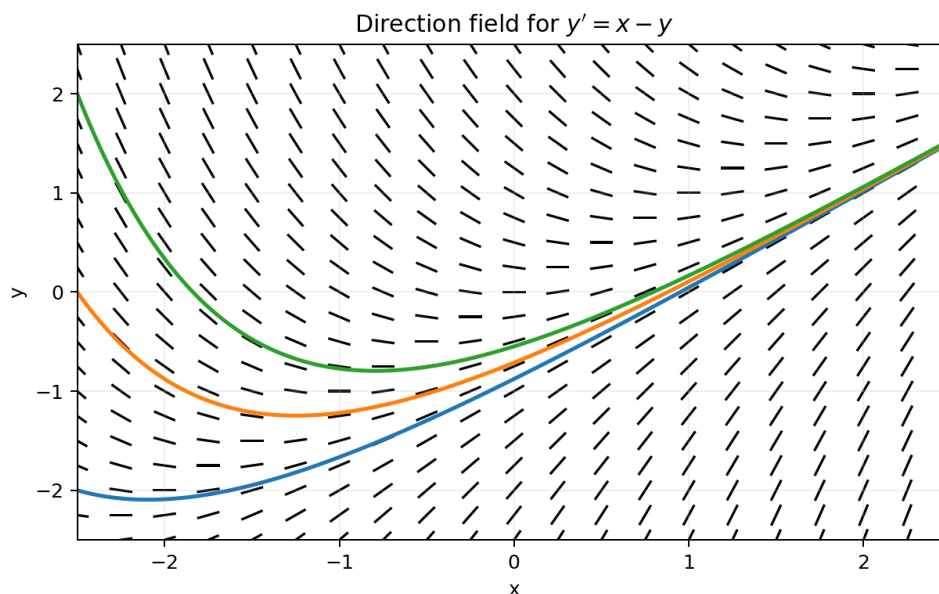


Figure 1: A direction field for  $y' = x - y$ . The differential equation prescribes a tangent direction at every point.

At  $(1, 2)$ , for example,

$$y' = 1 - 2 = -1,$$

so a solution passing through that point must have slope  $-1$  there.

#### Core idea

Before solving an ODE symbolically, ask four geometric questions: Where are the equilibria? Where is  $y' > 0$ ? Where is  $y' < 0$ ? Can solutions cross one another? These questions often reveal the qualitative behavior without any closed-form formula.

## 2 Separable and autonomous equations

### 2.1 Separation of variables

A separable equation has the form

$$\frac{dy}{dx} = f(x)g(y).$$

Formally one writes

$$\frac{dy}{g(y)} = f(x) dx,$$

and integrates.

#### Why separation works

Let  $G'(y) = 1/g(y)$  on a region where  $g(y) \neq 0$ . If  $y$  satisfies  $y' = f(x)g(y)$ , then

$$\frac{d}{dx}G(y(x)) = f(x).$$

Therefore

$$G(y(x)) = \int f(x) dx + C.$$

Proof

By the chain rule,

$$\frac{d}{dx}G(y(x)) = G'(y)y' = \frac{1}{g(y)}f(x)g(y) = f(x).$$

Integrating with respect to  $x$  proves the formula. Thus “moving  $dx$  and  $dy$ ” is shorthand for a chain-rule calculation, not ordinary fraction algebra.

Worked example

Solve

$$y' = xy.$$

Then

$$\frac{dy}{y} = x dx,$$

so

$$\ln |y| = \frac{x^2}{2} + C.$$

Exponentiating and absorbing the sign into the constant,

$$y = Ce^{x^2/2}.$$

## 2.2 Exponential growth and decay

For

$$y' = ky,$$

the solution is

$$y = Ce^{kt}.$$

If  $k > 0$ , the quantity grows exponentially; if  $k < 0$ , it decays exponentially. The doubling and half-life times are

$$T_{\text{double}} = \frac{\ln 2}{k} \quad (k > 0), \quad T_{1/2} = \frac{\ln 2}{|k|} \quad (k < 0).$$

## 2.3 The logistic equation

Unlimited exponential growth is often unrealistic. The logistic model is

$$P' = rP \left(1 - \frac{P}{K}\right),$$

where  $K$  is the carrying capacity.

Separating and using partial fractions gives

$$\frac{K}{P(K-P)} = \frac{1}{P} + \frac{1}{K-P},$$

so

$$\ln P - \ln(K-P) = rt + C.$$

Hence

$$P(t) = \frac{K}{1 + Be^{-rt}}.$$

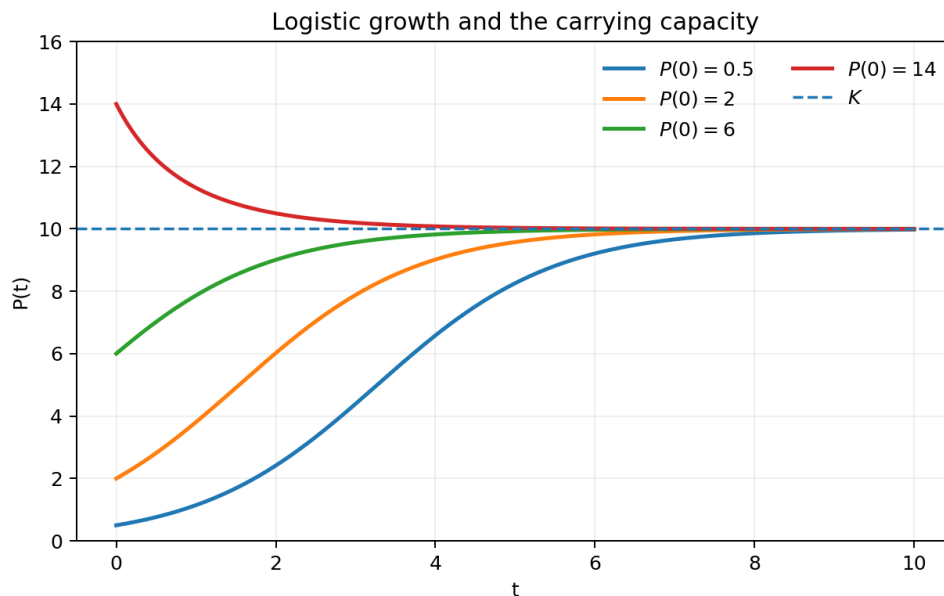


Figure 2: Logistic trajectories approach the stable carrying capacity  $K$ .

## 2.4 Autonomous equations and phase lines

An autonomous scalar equation has the form

$$y' = f(y).$$

Equilibria are values  $y_*$  satisfying  $f(y_*) = 0$ .

For

$$y' = y(1 - y),$$

we have equilibria 0 and 1. On  $0 < y < 1$ ,  $y' > 0$ ; for  $y > 1$ ,  $y' < 0$ . Hence  $y = 1$  attracts nearby positive solutions, while  $y = 0$  repels them.

### One-dimensional linear stability test

If  $f(y_*) = 0$  and  $f$  is differentiable near  $y_*$ , then:

- if  $f'(y_*) < 0$ , the equilibrium is locally asymptotically stable;
- if  $f'(y_*) > 0$ , it is unstable.

### Proof

Write  $y = y_* + \eta$ . Taylor expansion gives

$$f(y_* + \eta) = f(y_*) + f'(y_*)\eta + O(\eta^2) = f'(y_*)\eta + O(\eta^2).$$

For sufficiently small  $\eta$ , the dominant linear equation is

$$\eta' = f'(y_*)\eta,$$

whose solution behaves like  $\eta(t) \approx Ce^{f'(y_*)t}$ . Negative exponent means decay; positive exponent means growth.

### 3 First-order linear, exact, and Bernoulli equations

#### 3.1 The integrating-factor method

A first-order linear equation is

$$y' + p(x)y = q(x).$$

We seek a multiplier  $\mu(x)$  such that the left side becomes a product derivative. We want

$$(\mu y)' = \mu y' + \mu' y$$

to equal

$$\mu y' + \mu p y.$$

Therefore  $\mu' = p\mu$ , so

$$\mu(x) = e^{\int p(x) dx}.$$

Multiplying the ODE by  $\mu$  gives

$$(\mu y)' = \mu q,$$

and hence

$$y = \frac{1}{\mu} \left( \int \mu q dx + C \right).$$

#### Worked example

Solve

$$y' + 2y = e^x.$$

The integrating factor is  $\mu = e^{2x}$ . Therefore

$$(e^{2x}y)' = e^{3x},$$

so

$$e^{2x}y = \frac{1}{3}e^{3x} + C.$$

Thus

$$y = \frac{1}{3}e^x + Ce^{-2x}.$$

The particular solution  $\frac{1}{3}e^x$  responds to the forcing;  $Ce^{-2x}$  is the homogeneous transient.

#### 3.2 Exact differential equations

Consider

$$M(x, y) dx + N(x, y) dy = 0.$$

If there is a potential  $F(x, y)$  with

$$F_x = M, \quad F_y = N,$$

then

$$dF = M dx + N dy,$$

and every solution satisfies

$$F(x, y) = C.$$

#### Exactness criterion

If  $M, N$  have continuous first partial derivatives on a simply connected region, then the equation is exact precisely when

$$M_y = N_x.$$

#### Proof

Necessity follows from equality of mixed partials:

$$M_y = (F_x)_y = F_{xy} = F_{yx} = (F_y)_x = N_x.$$

For sufficiency, fix  $(x_0, y_0)$  and define

$$F(x, y) = \int_{x_0}^x M(s, y_0) ds + \int_{y_0}^y N(x, t) dt.$$

Differentiating and using  $M_y = N_x$  shows  $F_x = M$  and  $F_y = N$ .

#### Worked example

Solve

$$(2xy + 3) dx + (x^2 + 4y) dy = 0.$$

Since  $M_y = 2x = N_x$ , the equation is exact. Integrating  $F_x = 2xy + 3$  with respect to  $x$  gives

$$F = x^2y + 3x + g(y).$$

Then

$$F_y = x^2 + g'(y) = x^2 + 4y,$$

so  $g(y) = 2y^2$ . Therefore

$$x^2y + 3x + 2y^2 = C.$$

### 3.3 Bernoulli equations

A Bernoulli equation has the form

$$y' + p(x)y = q(x)y^n.$$

Set

$$v = y^{1-n}.$$

Then

$$v' = (1-n)y^{-n}y'.$$

After dividing the original equation by  $y^n$ , the equation becomes linear in  $v$ . This is a recurring mathematical theme: a nonlinear equation can become linear in the right coordinate.

## 4 Existence, uniqueness, and Picard iteration

### 4.1 Why “solving” starts before formulas

Given

$$y' = f(t, y), \quad y(t_0) = y_0,$$

there are three logically separate questions:

1. Does a solution exist?
2. Is it unique?
3. Can we express or approximate it effectively?

#### Picard-Lindelof theorem, local form

Suppose  $f$  is continuous in a rectangle around  $(t_0, y_0)$  and is Lipschitz in  $y$  there: for some  $L$ ,

$$|f(t, u) - f(t, v)| \leq L|u - v|.$$

Then the initial-value problem

$$y' = f(t, y), \quad y(t_0) = y_0$$

has a unique solution on a sufficiently small interval around  $t_0$ .

### 4.2 Integral formulation

The ODE is equivalent to

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) \, ds.$$

Indeed, integrate  $y' = f(t, y)$  from  $t_0$  to  $t$ ; conversely, differentiate the integral equation.

Picard iteration starts with a simple guess, often  $y_0(t) = y_0$ , and defines

$$y_{n+1}(t) = y_0 + \int_{t_0}^t f(s, y_n(s)) \, ds.$$

Under the theorem's hypotheses, this sequence converges to the unique solution.

### 4.3 A uniqueness proof via Gronwall

Assume  $y_1, y_2$  solve the same IVP. Then

$$y_1(t) - y_2(t) = \int_{t_0}^t [f(s, y_1(s)) - f(s, y_2(s))] \, ds.$$

Thus

$$|y_1(t) - y_2(t)| \leq L \int_{t_0}^t |y_1(s) - y_2(s)| \, ds.$$

**Gronwall's inequality, simple form**

If a nonnegative continuous function  $u$  satisfies

$$u(t) \leq A + B \int_{t_0}^t u(s) \, ds,$$

then

$$u(t) \leq Ae^{B(t-t_0)}.$$

In particular, if  $A = 0$ , then  $u \equiv 0$ .

Applying this with  $u = |y_1 - y_2|$  and  $A = 0$  yields  $y_1 = y_2$ .

**4.4 How uniqueness can fail**

Consider

$$y' = \sqrt{|y|}, \quad y(0) = 0.$$

One solution is  $y \equiv 0$ . But for every  $a \geq 0$ ,

$$y(t) = \begin{cases} 0, & t \leq a, \\ \frac{(t-a)^2}{4}, & t > a \end{cases}$$

is also a solution. The issue is that  $f(y) = \sqrt{|y|}$  is not Lipschitz near 0.

**Common mistake**

Continuity of  $f$  is enough for local existence under standard hypotheses, but it is not enough for uniqueness. The extra control in the  $y$ -direction matters.

**5 Second-order linear equations****5.1 Constant coefficients and the characteristic polynomial**

Consider

$$ay'' + by' + cy = 0.$$

Try an exponential

$$y = e^{rt}.$$

Then

$$y' = re^{rt}, \quad y'' = r^2e^{rt},$$

so

$$(ar^2 + br + c)e^{rt} = 0.$$

Since  $e^{rt} \neq 0$ ,

$$\boxed{ar^2 + br + c = 0}.$$

This is the characteristic equation.

## 5.2 Why exponentials are the natural trial functions

Differentiation maps  $e^{rt}$  to a scalar multiple of itself. In operator language, exponentials are eigenfunctions of the derivative operator:

$$De^{rt} = re^{rt}.$$

Therefore a polynomial in  $D$  becomes the same polynomial evaluated at  $r$ .

## 5.3 Distinct real roots

If the roots are  $r_1 \neq r_2$ , then

$$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}.$$

### Proof

Both exponentials satisfy the equation. They are linearly independent because

$$W(t) = \begin{vmatrix} e^{r_1 t} & e^{r_2 t} \\ r_1 e^{r_1 t} & r_2 e^{r_2 t} \end{vmatrix} = (r_2 - r_1) e^{(r_1 + r_2)t} \neq 0.$$

The solution space of a homogeneous second-order linear ODE is two-dimensional, so these two independent solutions span it.

### Worked example

For

$$y'' - 3y' + 2y = 0,$$

we solve

$$r^2 - 3r + 2 = (r - 1)(r - 2) = 0,$$

hence

$$y = C_1 e^t + C_2 e^{2t}.$$

## 5.4 Repeated roots

If  $r = r_0$  is a double root, the two independent solutions are

$$e^{r_0 t}, \quad t e^{r_0 t}.$$

Thus

$$y = (C_1 + C_2 t) e^{r_0 t}.$$

One way to understand  $t e^{rt}$  is to let two roots merge:

$$\lim_{\varepsilon \rightarrow 0} \frac{e^{(r+\varepsilon)t} - e^{rt}}{\varepsilon} = \frac{\partial}{\partial r} e^{rt} = t e^{rt}.$$

## 5.5 Complex roots and Euler's formula

If

$$r = \alpha \pm i\beta,$$

then

$$e^{(\alpha+i\beta)t} = e^{\alpha t} (\cos \beta t + i \sin \beta t).$$

Taking real and imaginary parts gives

$$y = e^{\alpha t} (C_1 \cos \beta t + C_2 \sin \beta t).$$

Complex roots therefore encode oscillation, while the real part  $\alpha$  controls exponential growth or decay.

## 6 Oscillation, damping, forcing, and resonance

### 6.1 The harmonic oscillator

Hooke's law is  $F = -kx$ , and Newton's law is  $F = mx''$ . Therefore

$$mx'' + kx = 0,$$

or

$$x'' + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{\frac{k}{m}}.$$

Hence

$$x(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t).$$

The period is

$$T = \frac{2\pi}{\omega_0}.$$

### 6.2 Energy conservation

Multiply

$$mx'' + kx = 0$$

by  $x'$ :

$$mx''x' + kxx' = 0.$$

But

$$mx''x' = \frac{d}{dt} \left( \frac{1}{2} m(x')^2 \right), \quad kxx' = \frac{d}{dt} \left( \frac{1}{2} kx^2 \right).$$

Therefore

$$\frac{d}{dt} \left( \frac{1}{2} m(x')^2 + \frac{1}{2} kx^2 \right) = 0,$$

so

$$E = \frac{1}{2} m(x')^2 + \frac{1}{2} kx^2$$

is conserved.

### 6.3 Damping

A viscous damping force proportional to velocity gives

$$mx'' + cx' + kx = 0.$$

The discriminant  $c^2 - 4mk$  separates three regimes:

- $c^2 > 4mk$ : overdamped, two negative real roots;
- $c^2 = 4mk$ : critically damped, repeated negative root;

- $c^2 < 4mk$ : underdamped, complex roots with negative real part.

In the underdamped case,

$$x(t) = e^{-ct/(2m)} (A \cos \omega_d t + B \sin \omega_d t).$$

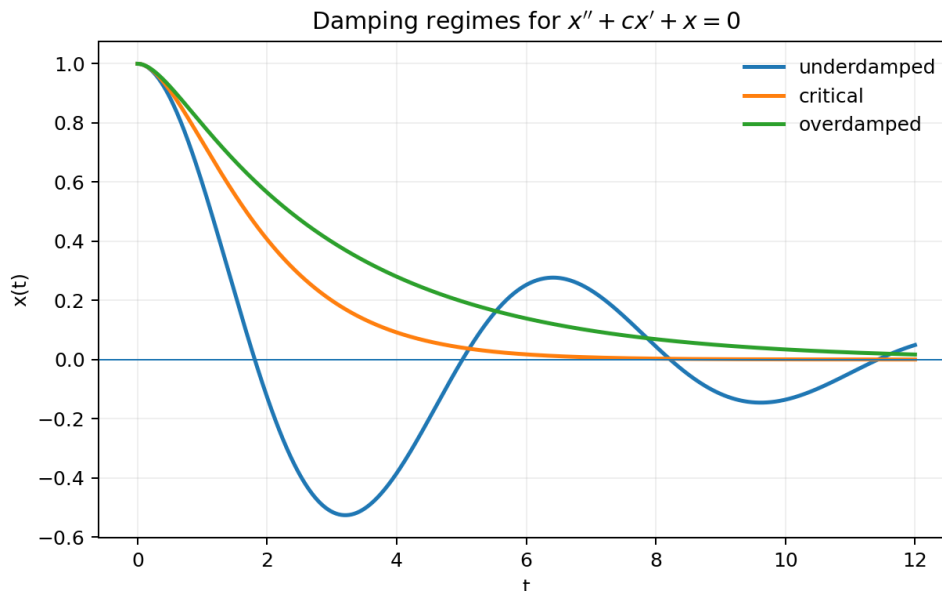


Figure 3: Underdamped, critical, and overdamped behavior.

## 6.4 Forced oscillation and resonance

With an external periodic force,

$$mx'' + cx' + kx = F_0 \cos \omega t.$$

The solution splits into

$$x = x_h + x_p,$$

where  $x_h$  is the transient and  $x_p$  the steady-state response.

For the ideal undamped system

$$x'' + \omega_0^2 x = F \cos \omega t,$$

when  $\omega \neq \omega_0$  one can take

$$x_p = \frac{F}{\omega_0^2 - \omega^2} \cos \omega t.$$

At exact resonance  $\omega = \omega_0$ , the forcing matches a homogeneous mode, and the correct particular solution is

$$x_p = \frac{F}{2\omega_0} t \sin \omega_0 t.$$

The factor  $t$  produces secular growth of amplitude.

## 7 Linear operators, superposition, and the Wronskian

## 7.1 Operator viewpoint

Define

$$L[y] = ay'' + by' + cy.$$

Linearity means

$$L[\alpha u + \beta v] = \alpha L[u] + \beta L[v].$$

Therefore if  $L[y_h] = 0$  and  $L[y_p] = g$ , then

$$\boxed{y = y_h + y_p}$$

is a solution of  $L[y] = g$ , and every solution has this form.

## 7.2 Undetermined coefficients

For

$$y'' - 3y' + 2y = e^{4t},$$

the homogeneous solution is

$$y_h = C_1 e^t + C_2 e^{2t}.$$

Try  $y_p = Ae^{4t}$ . Substitution yields  $6A = 1$ , so

$$\boxed{y = C_1 e^t + C_2 e^{2t} + \frac{1}{6} e^{4t}}.$$

If the forcing duplicates a homogeneous mode, multiply the guess by  $t$  enough times to obtain linear independence.

## 7.3 The Wronskian

For two functions,

$$\boxed{W(y_1, y_2) = y_1 y_2' - y_2 y_1'}.$$

If  $W(t_0) \neq 0$ , then  $y_1, y_2$  are linearly independent.

### Abel's identity

If  $y_1, y_2$  solve

$$y'' + p(t)y' + q(t)y = 0,$$

then

$$\boxed{W'(t) = -p(t)W(t)},$$

so

$$\boxed{W(t) = C e^{-\int p(t) dt}}.$$

### Proof

Differentiate

$$W = y_1 y_2' - y_2 y_1'.$$

Then

$$W' = y_1 y_2'' - y_2 y_1''.$$

Using  $y_i'' = -py_i' - qy_i$  gives

$$W' = -p(y_1y_2' - y_2y_1') = -pW.$$

The final formula follows by solving this first-order separable equation.

## 7.4 Variation of parameters

For

$$y'' + p(t)y' + q(t)y = g(t),$$

let  $y_1, y_2$  be a fundamental homogeneous pair. Seek

$$y_p = u_1(t)y_1(t) + u_2(t)y_2(t)$$

and impose

$$u_1'y_1 + u_2'y_2 = 0.$$

Substitution yields

$$u_1'y_1' + u_2'y_2' = g.$$

Solving the  $2 \times 2$  system gives

$$\boxed{u_1' = -\frac{y_2g}{W}}, \quad \boxed{u_2' = \frac{y_1g}{W}}.$$

This method applies far beyond the limited forcing families handled by undetermined coefficients.

## 8 Systems, eigenvalues, and matrix exponentials

### 8.1 Writing a system in matrix form

A linear system

$$\begin{cases} x' = 2x + y, \\ y' = x + 2y \end{cases}$$

becomes

$$\boxed{\mathbf{x}' = A\mathbf{x}}, \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

### 8.2 Why eigenvalues solve the system

Try

$$\mathbf{x}(t) = e^{\lambda t} \mathbf{v}.$$

Then

$$\lambda e^{\lambda t} \mathbf{v} = A e^{\lambda t} \mathbf{v},$$

so

$$\boxed{A\mathbf{v} = \lambda\mathbf{v}}.$$

Every eigenpair therefore gives a mode

$$\boxed{\mathbf{x}(t) = e^{\lambda t} \mathbf{v}}.$$

For the example matrix, the eigenpairs are

$$\lambda_1 = 3, \quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \lambda_2 = 1, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Hence

$$\mathbf{x}(t) = C_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^t \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

### 8.3 The matrix exponential

Define

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \cdots.$$

Termwise differentiation gives

$$\frac{d}{dt} e^{At} = A e^{At},$$

so the IVP

$$\mathbf{x}' = A\mathbf{x}, \quad \mathbf{x}(0) = \mathbf{x}_0$$

has solution

$$\mathbf{x}(t) = e^{At} \mathbf{x}_0.$$

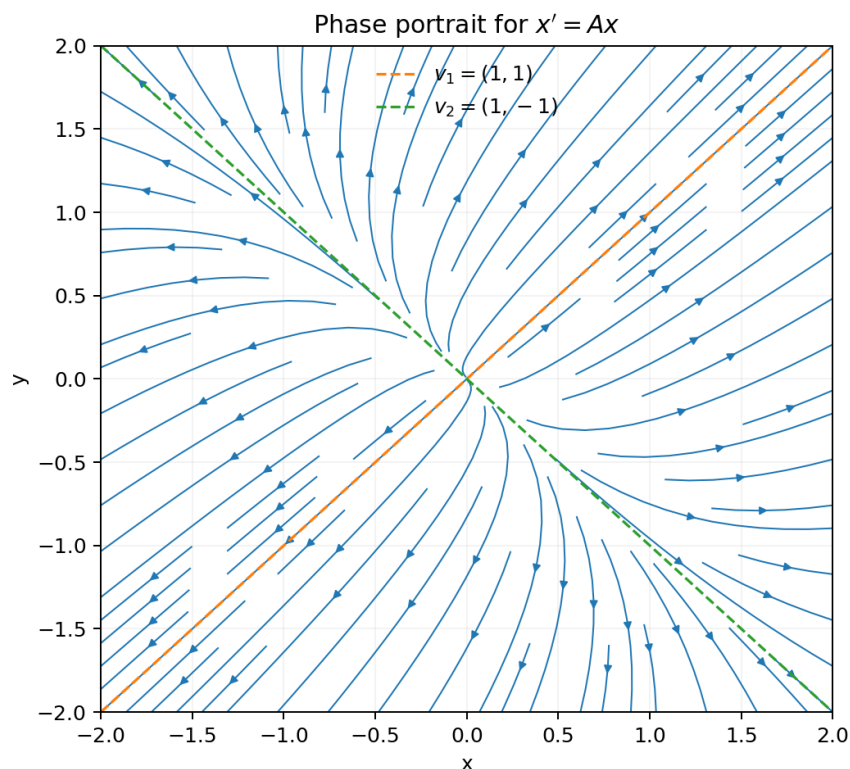


Figure 4: A phase portrait. Eigenvectors determine invariant directions; eigenvalues determine growth or decay along them.

### 8.4 Classification in two dimensions

For a  $2 \times 2$  system near the origin:

- two negative real eigenvalues: stable node;
- two positive real eigenvalues: unstable node;
- opposite signs: saddle;
- $\alpha \pm i\beta$ ,  $\alpha < 0$ : stable spiral;
- $\alpha \pm i\beta$ ,  $\alpha > 0$ : unstable spiral;
- purely imaginary eigenvalues: linear center.

#### Core idea

Eigenvalues are not merely algebraic numbers. In dynamics, the real part tells you whether a mode grows or decays; the imaginary part tells you whether it rotates or oscillates.

## 9 Nonlinear systems and stability

### 9.1 Linearization

For

$$\mathbf{x}' = \mathbf{f}(\mathbf{x}),$$

an equilibrium  $\mathbf{x}_*$  satisfies  $\mathbf{f}(\mathbf{x}_*) = 0$ . Write

$$\mathbf{x} = \mathbf{x}_* + \mathbf{u}.$$

Taylor expansion gives

$$\mathbf{f}(\mathbf{x}_* + \mathbf{u}) = J(\mathbf{x}_*)\mathbf{u} + O(\|\mathbf{u}\|^2),$$

where  $J$  is the Jacobian. Thus near the equilibrium,

$$\mathbf{u}' \approx J(\mathbf{x}_*)\mathbf{u}.$$

The eigenvalues of the Jacobian often determine local behavior.

### 9.2 Lyapunov functions

A Lyapunov function is an energy-like scalar  $V(\mathbf{x})$  satisfying  $V \geq 0$  and  $V(\mathbf{x}_*) = 0$ . If along trajectories

$$\dot{V} = \nabla V \cdot \mathbf{f}(\mathbf{x}) < 0$$

away from equilibrium, then the “energy” decreases.

For the damped oscillator

$$mx'' + cx' + kx = 0,$$

take

$$V = \frac{1}{2}m(x')^2 + \frac{1}{2}kx^2.$$

Then

$$\dot{V} = x'(mx'' + kx) = -c(x')^2 \leq 0.$$

Thus damping continuously removes mechanical energy.

## Lyapunov stability principle, informal form

If  $V$  is positive definite near an equilibrium and  $\dot{V}$  is negative definite there, the equilibrium is asymptotically stable. If  $\dot{V} \leq 0$ , one obtains stability under additional standard conditions.

## 9.3 What linearization cannot decide

If the Jacobian has eigenvalues with zero real part, linearization may be inconclusive because nonlinear terms can dominate. This is where Lyapunov functions, invariant sets, normal forms, or deeper dynamical-systems tools become necessary.

## 10 Laplace transforms and input-output thinking

## 10.1 Definition

For suitable  $f(t)$ ,

$$\mathcal{L}\{f\}(s) = F(s) = \int_0^\infty e^{-st} f(t) dt.$$

Useful transforms include

$$\begin{aligned} \mathcal{L}\{1\} &= \frac{1}{s}, & \mathcal{L}\{e^{at}\} &= \frac{1}{s-a}, \\ \mathcal{L}\{\cos \omega t\} &= \frac{s}{s^2 + \omega^2}, & \mathcal{L}\{\sin \omega t\} &= \frac{\omega}{s^2 + \omega^2}. \end{aligned}$$

## 10.2 Transform of derivatives

## Theorem

If  $F(s) = \mathcal{L}\{f\}$  and the boundary terms are well behaved, then

$$\mathcal{L}\{f'\} = sF(s) - f(0).$$

Similarly,

$$\mathcal{L}\{f''\} = s^2 F(s) - sf(0) - f'(0).$$

## Proof

Integration by parts gives

$$\int_0^\infty e^{-st} f'(t) dt = \left[ e^{-st} f(t) \right]_0^\infty + s \int_0^\infty e^{-st} f(t) dt.$$

Assuming the term at infinity vanishes, the boundary contribution is  $-f(0)$ , proving the formula.

## Worked example

Solve

$$y'' + y = 0, \quad y(0) = 0, \quad y'(0) = 1.$$

Taking transforms,

$$s^2 Y - 1 + Y = 0,$$

so

$$Y = \frac{1}{s^2 + 1}.$$

Therefore

$$y(t) = \sin t.$$

### 10.3 Step functions and time shifts

The Heaviside step function is

$$H(t - a) = \begin{cases} 0, & t < a, \\ 1, & t \geq a. \end{cases}$$

If  $G(s) = \mathcal{L}\{g(t)\}$ , then

$$\mathcal{L}\{H(t - a)g(t - a)\} = e^{-as}G(s).$$

This is ideal for switched-on forces, piecewise inputs, and control systems.

### 10.4 Impulses

The Dirac delta is characterized by

$$\int f(t)\delta(t - a) dt = f(a).$$

Its Laplace transform is

$$\mathcal{L}\{\delta(t - a)\} = e^{-as}.$$

It models an idealized instantaneous impulse.

### 10.5 Convolution

Define

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau.$$

Then

$$\mathcal{L}\{f * g\} = F(s)G(s).$$

This is the mathematical basis of impulse-response representations: a linear time-invariant system's response to a general input can be built by superposing shifted impulse responses.

## 11 Power series and numerical methods

### 11.1 Power-series solutions

When elementary functions fail, try

$$y = \sum_{n=0}^{\infty} a_n x^n.$$

Then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Substitute into the ODE and equate coefficients.

## Worked example

For

$$y'' + y = 0,$$

re-indexing gives

$$(n+2)(n+1)a_{n+2} + a_n = 0,$$

so

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)}.$$

The even terms form

$$a_0 \left( 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \right) = a_0 \cos x,$$

and the odd terms form

$$a_1 \left( x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \right) = a_1 \sin x.$$

Thus the series method reconstructs the familiar solution  $a_0 \cos x + a_1 \sin x$ .

## 11.2 Euler's method

For

$$y' = f(t, y),$$

Taylor's theorem gives

$$y(t+h) = y(t) + hy'(t) + O(h^2).$$

Replacing  $y'$  by  $f(t, y)$  and discarding higher-order terms yields

$$y_{n+1} = y_n + hf(t_n, y_n).$$

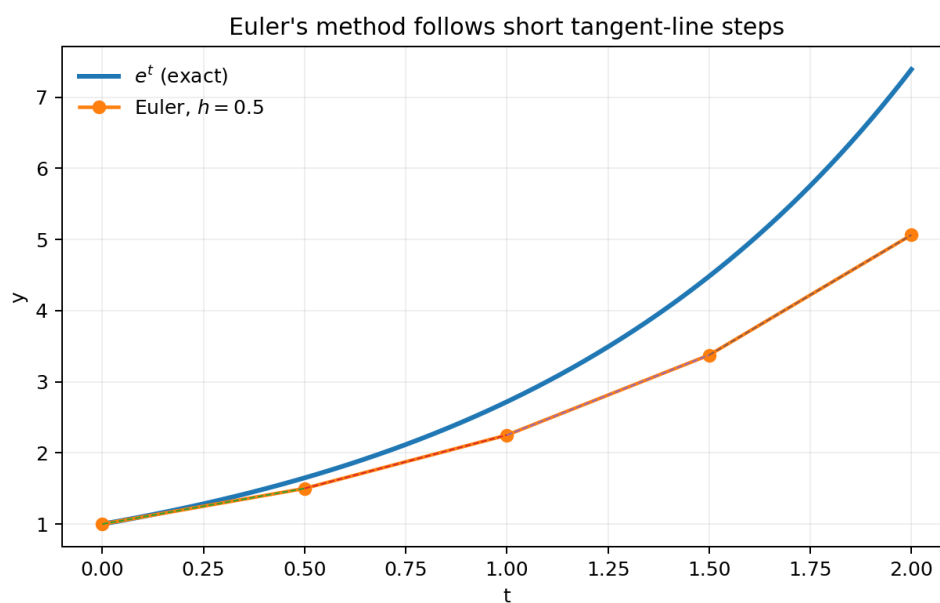


Figure 5: Euler's method repeatedly follows a tangent line for a short step.

The local truncation error is  $O(h^2)$ ; over  $O(1/h)$  steps, the global error is typically  $O(h)$ .

### 11.3 Stability of Euler's method

Apply Euler to

$$y' = \lambda y, \quad \lambda < 0.$$

Then

$$y_{n+1} = (1 + h\lambda)y_n,$$

so

$$y_n = (1 + h\lambda)^n y_0.$$

The numerical solution decays only when

$$|1 + h\lambda| < 1,$$

which for real negative  $\lambda$  means

$$0 < h < \frac{2}{|\lambda|}.$$

Thus a stable physical system can be made numerically unstable by choosing a step size that is too large.

### 11.4 Runge-Kutta 4

RK4 samples four slopes:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \\ k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \\ k_4 &= f(t_n + h, y_n + hk_3), \end{aligned}$$

then updates

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Its global error is  $O(h^4)$  under standard smoothness assumptions.

## 12 Boundary-value problems and eigenfunctions

### 12.1 From initial data to boundary data

Consider

$$y'' + \lambda y = 0, \quad y(0) = 0, \quad y(L) = 0.$$

For  $\lambda > 0$ ,

$$y = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x).$$

The first boundary condition gives  $A = 0$ . The second requires

$$\sin(\sqrt{\lambda}L) = 0.$$

Therefore

$$\sqrt{\lambda}L = n\pi,$$

so

$$\lambda_n = \frac{n^2\pi^2}{L^2}.$$

Only special values of  $\lambda$  permit nonzero solutions. These are eigenvalues, with eigenfunctions

$$y_n(x) = \sin \frac{n\pi x}{L}.$$

## 12.2 Sturm-Liouville form

A major class of problems is

$$-\frac{d}{dx}(p(x)y') + q(x)y = \lambda w(x)y.$$

Under suitable boundary conditions, the operator is self-adjoint. Its eigenvalues are real, eigenfunctions for distinct eigenvalues are orthogonal with weight  $w$ , and under standard hypotheses the eigenfunctions form a basis.

### Orthogonality of distinct eigenfunctions

If

$$L[y_n] = \lambda_n w y_n, \quad L[y_m] = \lambda_m w y_m,$$

and  $L$  is self-adjoint with compatible boundary conditions, then for  $\lambda_n \neq \lambda_m$ ,

$$\int_a^b w(x) y_n(x) y_m(x) dx = 0.$$

### Proof

Multiply the first equation by  $y_m$ , the second by  $y_n$ , subtract, and integrate. Self-adjointness plus the boundary conditions cancel the derivative terms, leaving

$$(\lambda_n - \lambda_m) \int_a^b w y_n y_m dx = 0.$$

Since the eigenvalues differ, the integral must vanish.

## 13 Fourier series and the bridge to PDEs

### 13.1 Fourier series as coordinates in function space

On  $[-\pi, \pi]$ , a suitable function can be expanded as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

The orthogonality relations imply

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx,$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx.$$

This is the infinite-dimensional analogue of decomposing a vector into orthogonal basis vectors.

### 13.2 The heat equation

Let  $u(x, t)$  be the temperature of a thin rod. Fourier's law says heat flux is

$$q = -ku_x.$$

Conservation of energy on a small interval gives

$$\rho c u_t = -q_x = ku_{xx}.$$

Hence

$$\boxed{u_t = \alpha u_{xx}}, \quad \alpha = \frac{k}{\rho c}.$$

### 13.3 Separation of variables for the heat equation

With  $u(0, t) = u(L, t) = 0$ , try

$$u(x, t) = X(x)T(t).$$

Then

$$XT' = \alpha X''T,$$

so

$$\frac{T'}{\alpha T} = \frac{X''}{X} = -\lambda.$$

Thus

$$X'' + \lambda X = 0, \quad T' + \alpha \lambda T = 0.$$

The boundary conditions give

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n = \sin \frac{n\pi x}{L},$$

and

$$T_n = e^{-\alpha(n\pi/L)^2 t}.$$

Therefore

$$\boxed{u(x, t) = \sum_{n=1}^{\infty} b_n e^{-\alpha(n\pi/L)^2 t} \sin \frac{n\pi x}{L}}.$$

High-frequency modes have larger  $n^2$ , so they decay faster. That is the mathematical reason heat diffusion smooths sharp spatial features.

### 13.4 The wave equation

A vibrating string satisfies

$$\boxed{u_{tt} = c^2 u_{xx}}.$$

Separation gives spatial sine modes and temporal harmonic oscillators. With fixed ends,

$$\boxed{u(x, t) = \sum_{n=1}^{\infty} \left[ A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right] \sin \frac{n\pi x}{L}}.$$

Each integer  $n$  is a normal mode of vibration.

## 14 Unifying perspective

### Core idea

A large part of differential equations can be understood through one recurring structure: find modes that are preserved by the differential operator, then combine those modes to satisfy data and forcing.

For scalar constant-coefficient ODEs,

$$y = e^{\lambda t}$$

is a mode because differentiation preserves exponentials.

For systems,

$$\mathbf{x} = e^{\lambda t} \mathbf{v}$$

is a mode precisely when

$$A\mathbf{v} = \lambda\mathbf{v}.$$

For boundary-value problems,

$$L[\phi] = \lambda\phi$$

defines eigenfunctions. Fourier series are expansions in these eigenmodes. PDE solutions are then assembled from spatial eigenfunctions multiplied by time-dependent ODE solutions.

The conceptual chain is

$$\text{ODEs} \longrightarrow \text{linear algebra} \longrightarrow \text{eigenvalues} \longrightarrow \text{Fourier analysis} \longrightarrow \text{PDEs}.$$

## 15 Problem-solving playbook

When facing a differential equation, use the following decision process.

1. **Classify it.** What is the order? Linear or nonlinear? Autonomous? Constant coefficients? A system?
2. **Inspect geometry.** Find equilibria and signs before manipulating formulas.
3. **Check for a direct first-order structure.**
  - separable:  $y' = f(x)g(y)$ ;
  - linear:  $y' + py = q$ ;
  - exact:  $M dx + N dy = 0$  with  $M_y = N_x$ ;
  - Bernoulli:  $y' + py = qy^n$ .
4. **For constant-coefficient linear ODEs,** form the characteristic polynomial.
5. **For forced linear equations,** solve homogeneous + particular.
6. **For systems,** compute eigenvalues/eigenvectors or  $e^{At}$ .
7. **For discontinuous inputs or IVPs,** consider Laplace transforms.
8. **When no closed form is available,** use qualitative analysis, series, or numerical methods.
9. **Always verify.** Substitute your result into the original equation and check initial or boundary conditions.

## 16 Formula and theorem reference

### First order

Separable:

$$y' = f(x)g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x) dx + C.$$

Linear:

$$y' + py = q, \quad \mu = e^{\int p dx}, \quad y = \mu^{-1} \left( \int \mu q dx + C \right)^{\text{Euler:}}$$

Exact:

$$M dx + N dy = 0, \quad M_y = N_x, \quad F(x, y) = C.$$

### Second order

$$ay'' + by' + cy = 0 \Rightarrow ar^2 + br + c = 0.$$

Distinct roots:

$$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}.$$

Repeated root:

$$y = (C_1 + C_2 t) e^{rt}.$$

Complex roots:

$$y = e^{\alpha t} (C_1 \cos \beta t + C_2 \sin \beta t).$$

### Systems

$$\mathbf{x}' = A\mathbf{x}, \quad A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{x} = e^{\lambda t}\mathbf{v}.$$

$$\mathbf{x}(t) = e^{At}\mathbf{x}_0.$$

### Laplace

$$\mathcal{L}\{f'\} = sF - f(0),$$

$$\mathcal{L}\{f''\} = s^2 F - sf(0) - f'(0).$$

$$\mathcal{L}\{H(t-a)g(t-a)\} = e^{-as}G(s).$$

$$\mathcal{L}\{f * g\} = FG.$$

### Numerics

$$y_{n+1} = y_n + hf(t_n, y_n).$$

Euler stability for  $y' = \lambda y$ ,  $\lambda < 0$ :

$$0 < h < \frac{2}{|\lambda|}.$$

RK4:

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

### Stability

Scalar equilibrium  $f(y_*) = 0$ :

$$f'(y_*) < 0 \Rightarrow \text{stable}, \quad f'(y_*) > 0 \Rightarrow \text{unstable}.$$

Lyapunov:

$$V > 0, \quad \dot{V} < 0 \Rightarrow \text{asymptotic stability (locally)}.$$

### Boundary modes

Fixed-end eigenproblem:

$$y'' + \lambda y = 0, \quad y(0) = y(L) = 0,$$

$$\lambda_n = \left( \frac{n\pi}{L} \right)^2, \quad y_n = \sin \frac{n\pi x}{L}.$$

## 17 Suggested mastery exercises

These are ordered to build technique and proof-level understanding.

1. Solve  $y' = x(1 + y^2)$  with  $y(0) = 0$ .
2. Analyze equilibria and stability for  $y' = y(y-1)(2-y)$ .
3. Solve  $y' - 3y = x^2$  using an integrating factor.
4. Determine whether  $(3x^2y + 2)dx + (x^3 + 4y)dy = 0$  is exact, then solve it.
5. Prove that two solutions of a first-order IVP cannot cross when the hypotheses of uniqueness hold.

6. Solve  $y'' + 4y' + 13y = 0$  and interpret the damping/oscillation.
7. Solve  $y'' + y = \cos t$  and explain why resonance changes the trial particular solution.
8. Compute the Wronskian of  $e^{2t}$  and  $te^{2t}$ .
9. Solve  $\mathbf{x}' = A\mathbf{x}$  for  $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and interpret the phase portrait.
10. Linearize the nonlinear system  $x' = x - x^3 - y$ ,  $y' = x + y - y^3$  at each equilibrium you can find.
11. Use a Lyapunov energy to prove dissipation for  $x'' + 2x' + 5x = 0$ .
12. Solve  $y'' + 4y = H(t - \pi)$  with zero initial data using Laplace transforms.
13. Derive Euler's method directly from Taylor's theorem and explain local versus global error.
14. Apply Euler to  $y' = -10y$  with  $h = 0.3$  and explain the numerical instability.
15. Derive the fixed-end eigenvalues for  $X'' + \lambda X = 0$  on  $[0, L]$ .
16. Derive the separated solutions of the heat equation and explain why higher-frequency modes decay faster.

## Final perspective

The most important skill is not recognizing a formula; it is recognizing structure. A differential equation is a rule for tangent directions, acceleration, flux, or state evolution. Exact methods exploit special structures. Existence and uniqueness theorems explain when the rule actually determines a trajectory. Eigenvalues reveal natural modes. Transforms reorganize derivatives into algebra. Numerical methods approximate motion when symbolic formulas fail. Fourier analysis and eigenfunctions extend the same logic from finite-dimensional systems to functions and PDEs.

If these connections are clear, differential equations stop looking like a bag of tricks and start looking like one coherent theory of how local laws generate global behavior.